Part 4: Simple Linear Regression Theory

This lecture provides an overview of the most basic regression model, the **simple linear regression model**. There is a single **covariate/predcitor** x used to predict the **response** Y. This model will serve as a building block for the more complex (and more useful) models we will present afterwards.

The simple linear regression model is typically written using notation

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \ldots, n$$

where β_0 is the **intercept** and β_1 is the **slope** for the regression line.

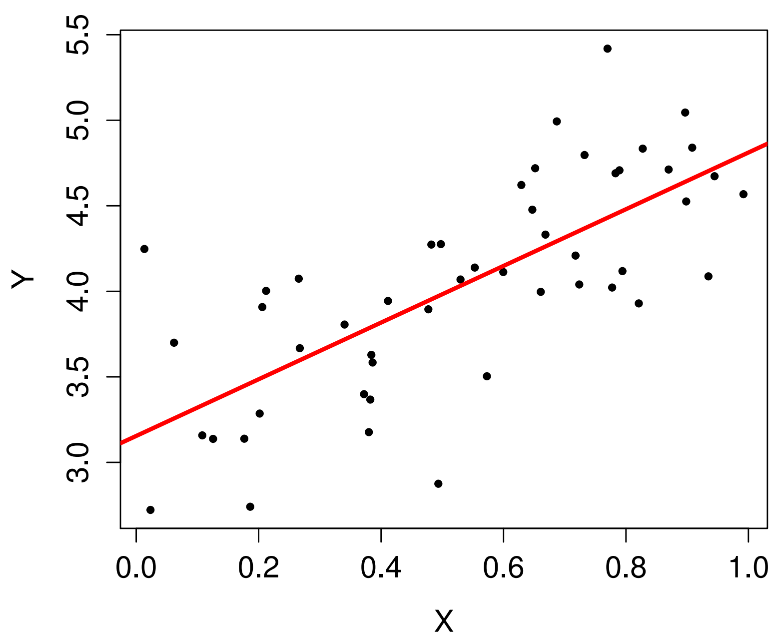


Figure 1: Synthetic scatter plot with $n = 50$ and a fitted regression line, $Y = 3.15 + 1.65x$.

The ϵ_i are random variables capturing the **irreducible error**, the scatter around the regression line.

The exact assumptions placed on the ϵ_i will vary depending on the situation, but the base assumptions that are almost always associated with simple linear regression are as follows:

1. $E(\epsilon_i) = 0 \quad$ for all i
2. $V(\epsilon_i) = \sigma^2 \quad$ for all i
3. The ϵ_i are uncorrelated, i.e., $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$.

Additional assumptions, **if justified**, lead to stronger results. In particular, one often assumes that the ϵ_i are normally distributed.

Exercise: Discuss the practical implications of these assumptions.

Example: This model is sometimes written as

$$Y = \alpha + \beta x + \epsilon.$$

This notation is the origin for references to the "beta" for a stock.

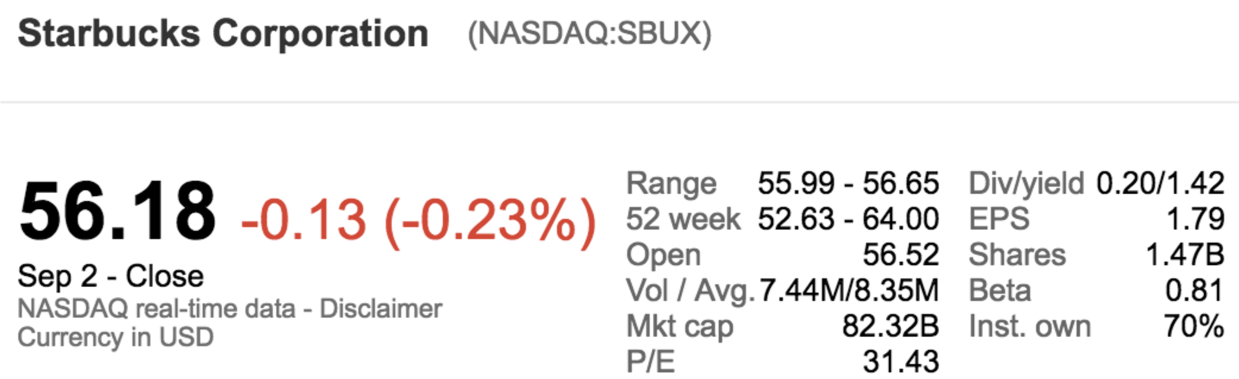


Figure 2: Screen shot from Google Finance, taken September 3, 2016.

In particular, the "beta" is found my performing a simple linear regression where the response is the excess returns for that equity, and the predictor is the excess returns for the market (as measured by some standard index such as S & P 500).

Figure 3 shows an example of the use of simple linear regression with the objective of calculating the "beta" for Google (GOOG), using daily market from 2014.

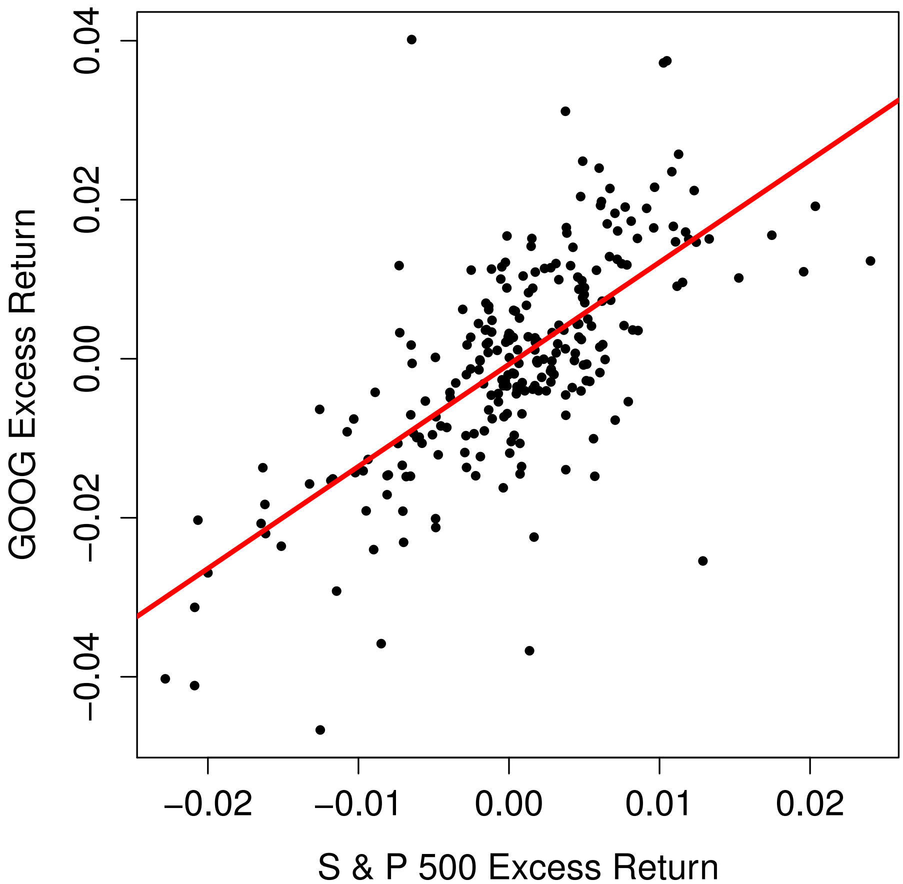


Figure 3: Daily excess returns for GOOG versus daily excess returns for S & P 500 for 2014.

Estimation via a Sample

In practice, we rely on a sample of n pairs (x_i, y_i) for $i = 1, 2, \ldots, n$, assumed to be drawn from a larger population. These will be used to estimate β_0, β_1 and σ^2. The standard notation for these estimates will be $\widehat{\beta}_0$, $\widehat{\beta}_1$, and $\widehat{\sigma}^2$.

The sample is naturally depicted in a **scatter plot**, with example shown in Figure 4.

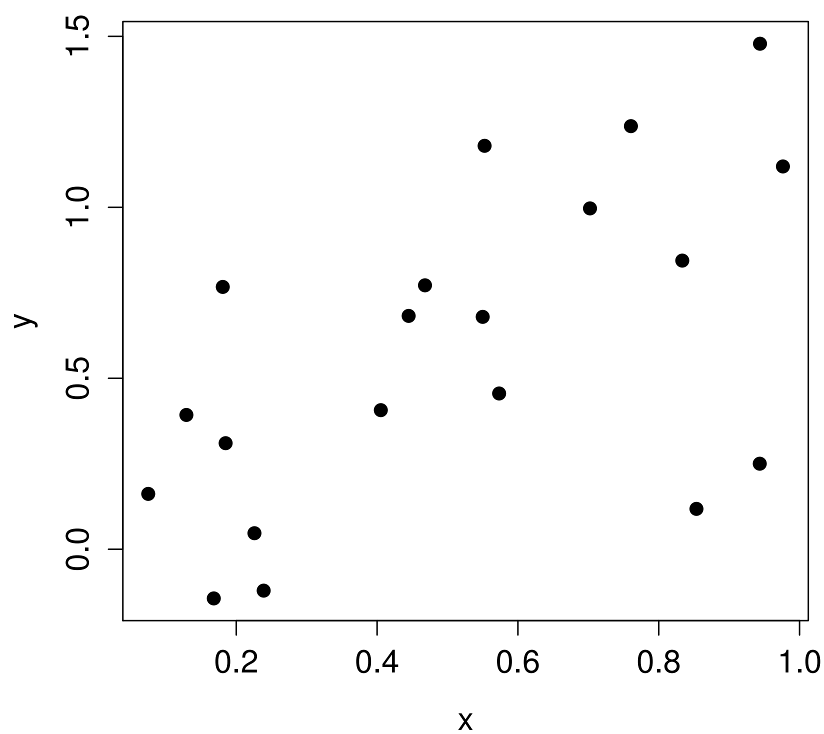


Figure 4: A simple scatter plot, with $n = 20$.

Some additional notation we should establish immediately: For any estimated regression line, the **fitted values** are

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i, \quad i = 1, 2, \ldots, n$$

and the **residuals** are

$$\widehat{\epsilon}_i = y_i - \widehat{y}_i, \quad i = 1, 2, \ldots, n.$$

Exercise: Use the scatter plot below, with a fit regression line superimposed, to demonstrate the fitted values and the residuals.

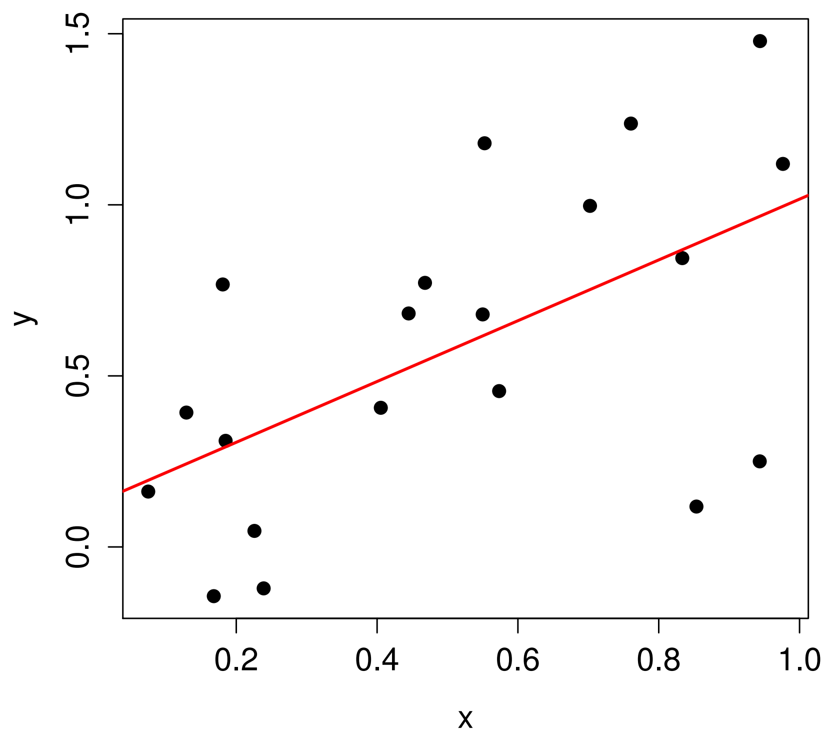


Figure 5: A simple scatter plot, with $n = 20$, with a regression line fit shown.

The Sample Covariance and Correlation

⋆ **Exercise:** Contrast covariance and correlation.

Recall that the covariance (and a related quantity, correlation) are measures of the **linear** association between two random variables X and Y:

$$\mathrm{Cov}(X,Y) = \mathrm{E}[(X - \mathrm{E}(X))(Y - \mathrm{E}(Y))]$$

and

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Not surprisingly, sample versions of these quantities are relevant to simple linear regression.

First, the **sample covariance** between observations of n pairs (x_i, y_i):

$$s_{xy} = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})$$

Note the similarity between this and the usual sample variance:

$$s_x^2 = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} (x_i - \overline{x})^2$$

Then, the **sample correlation** is

$$r_{xy} = \frac{s_{xy}}{s_x s_y}.$$

This quantity is typically just denoted with r. Figure 6 shows some examples of scatter plots, and the value of r for each. It is useful for building some intuition as to what different values of r mean.

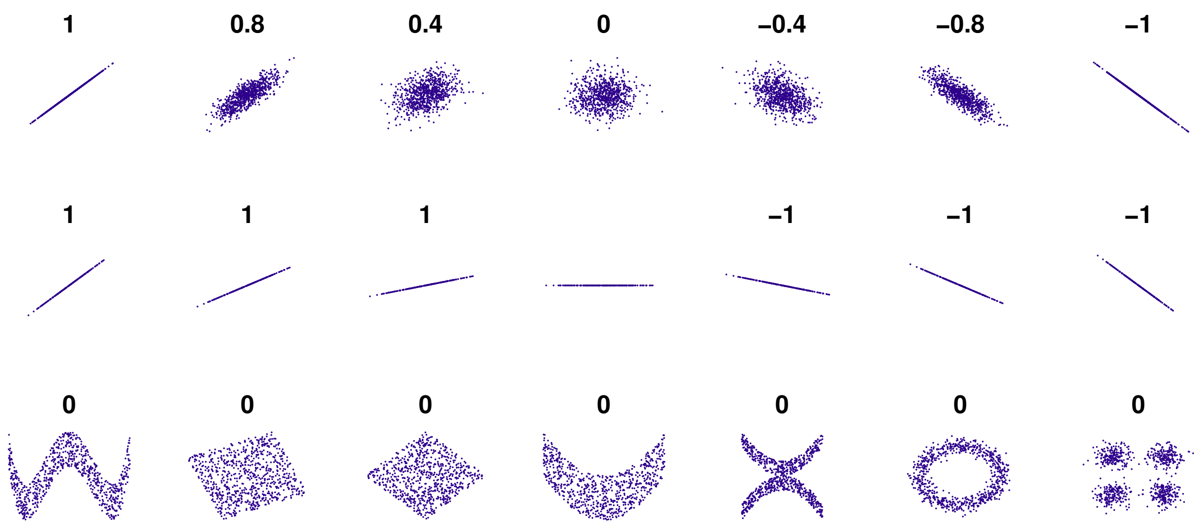
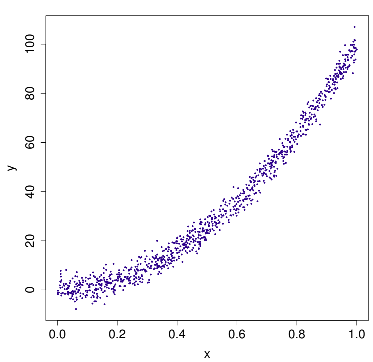


Figure 6: Examples of scatter plots, and corresponding values for r.

Exercise: Consider the scatter plot below. What, approximately, is the sample correlation?



The Parameter Estimates

⋆ **Exercise:** What route would you use to derive $\widehat{\beta}_0$ and $\widehat{\beta}_1$?

If we were to follow through with this, we would find that the optimal estimates would result from minimizing the **residual sum of squares (RSS)**:

$$\mathrm{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^2$$

And a little bit of calculus would show that

$$\widehat{\beta}_1 = \frac{\sum_i \left(y_i - \overline{y}\right)\left(x_i - \overline{x}\right)}{\sum_i \left(x_i - \overline{x}\right)^2} = \frac{s_{xy}}{s_x^2} = r_{xy}\left(\frac{s_y}{s_x}\right)$$

and

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}.$$

⋆ **Exercise:** Why do we use **least squares** to estimate the parameters in regression?

⋆ **Exercise:** Suppose that the roles of the predictor and response variables were interchanged in this regression model. What would be the effect on the slope of the regression line?

Figure 7 illustrates this idea. On the same scatter plot, two regression lines are shown: The "dashed-dotted" line is the regression line for predicting GOOG excess return from the S&P 500 excess return. The "dashed" line is the regression line for predicting S&P 500 excess return from GOOG excess return.

The solid line is the **SD Line.** This has a slope equal to s_y/s_x.

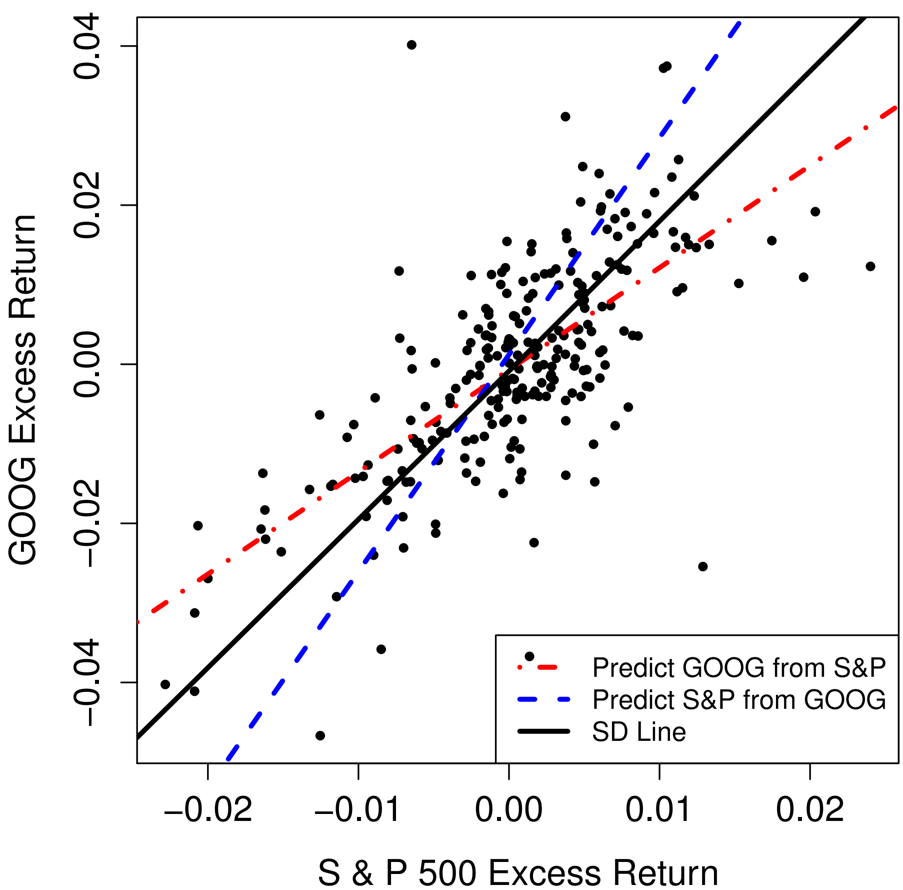


Figure 7: Daily excess returns for GOOG versus daily excess returns for S & P 500 for Jan. to Nov. 2012.

Estimating σ^2

The MLE for σ^2 can be derived to be

$$\widehat{\sigma}^2_{\text{MLE}} = \left(\frac{1}{n}\right) \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 = \text{RSS}\big/ n.$$

This estimator is biased, but it can be adjusted to obtain an unbiased estimator

$$\widehat{\sigma}^2 = \left(\frac{n}{n-2}\right) \sigma^2_{\text{MLE}} = \left(\frac{1}{n-2}\right) \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 = \text{RSS}\big/ (n-2).$$

This is the estimator we will typically use, and what is reported by R and other software packages.

Exercise: Is there intuition as to why we divide by $(n-2)$?

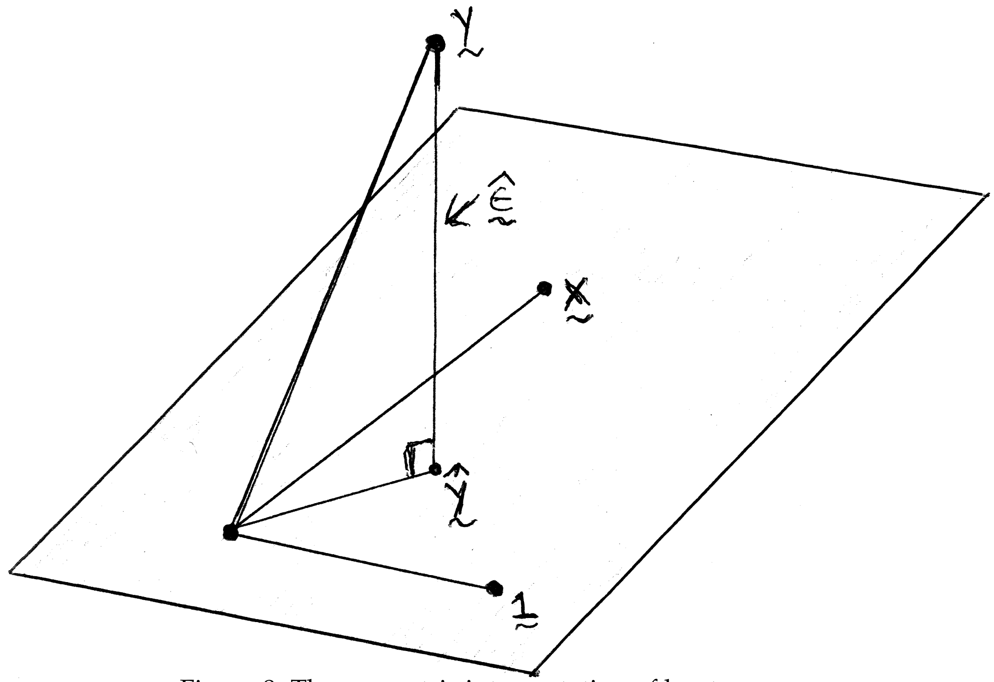


Figure 8: The geometric interpretation of least squares.

Figure 8 is a very useful geometric depiction of what happens when we do least squares. The vector of fitted values $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the space spanned by the vectors $\mathbf{1}$ and $\mathbf{x}$. (The rectangle represents all of the possible vectors that can be formed from linear combinations of $\mathbf{1}$ and $\mathbf{x}$.)

⋆ **Exercise:** What is the sum of the residuals in OLS?

Inference Regarding β_1

Within the context of this simple model, the primary objective is to perform inference regarding β_1, since it captures information regarding the relationship (or lack thereof) between the response and the single predictor.

Here are some key results in this direction. Keep in mind that these are all true under the assumptions put forth for the simple linear regression model.

- $\widehat{\beta}_1$ is an unbiased estimator of β_1.
- The variance of $\widehat{\beta}_1$ is
$$V(\widehat{\beta}_1) = \frac{\sigma^2}{(n-1)s_x^2} \approx \frac{\widehat{\sigma}^2}{(n-1)s_x^2}.$$
This means that the standard error is
$$\mathrm{SE}(\widehat{\beta}_1) \approx \frac{\widehat{\sigma}}{s_x\sqrt{(n-1)}}.$$
- Suppose that the true value of the slope is β_{10}. Then
$$T = \frac{\widehat{\beta}_1 - \beta_{10}}{\mathrm{SE}(\widehat{\beta}_1)}$$
has the t-distribution with $(n-2)$ degrees of freedom. Of course, when n is sufficiently large, T will be approximately standard normal.

Exercise: Construct a $100(1-\alpha)\%$ confidence interval for β_1.

⋆ **Exercise:** Construct a level α hypothesis tests regarding β_1. Consider both the one-sided and two-sided alternatives.

A common (overused) objective is to test for evidence if one can claim that $\beta_1 \neq 0$.

Exercise: Give two ways of using the inference procedures constructed above to address this objective.

It is important to be mindful of the following fact: If you fail to reject $H_0 : \beta_1 = 0$, or if the confidence interval for β_1 includes zero, this is often not because $\beta_1 = 0$, but instead because the standard error for $\widehat{\beta}_1$ is too large to detect the deviation from zero.

⋆ **Exercise:** How can the standard error of $\widehat{\beta}_1$ be decreased?

One of the most common mistakes made when constructing models is to blindly remove a predictor from the model solely because there is not strong evidence that its cofficient does not equal zero. Model building (the process of deciding which covariates to include) should comprise many factors, including our knowledge of the situation, the importance that we are placing on constructing a **parsimonious** model, **and** the results of formal statistical inference procedures.

The Coefficient of Determination, R^2

When working with regression models it is very common to hear references to "the R^2 for the model."

This is also called the **coefficient of determination**.

The idea is simple: R^2 equals the proportion of the sum of squared response which is accounted for by the model that has been fit, relative to the model with no covariates.

In particular:

$$R^2 = 1 - \left[\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 \Bigg/ \sum_{i=1}^{n} (y_i - \overline{y})^2 \right]$$

Note that $0 \leq R^2 \leq 1$.

In the case of simple linear regression (a single predictor), R^2 does indeed equal the sample correlation (r) squared.

R^2 is overused. In particular, note that it is possible for R^2 to be close to one, but the model be a terrible fit. And it is possible for a great-fitting model to have R^2 close to zero. See Figure 9.

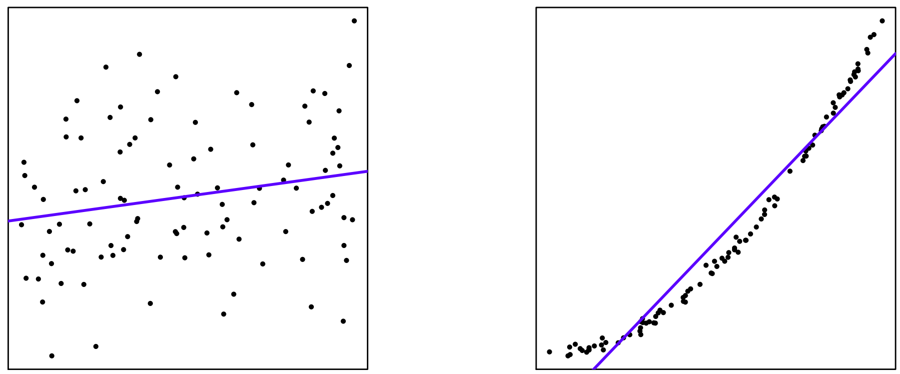


Figure 9: The scatter plot on the left has $R^2 \approx 0.05$, while that on the right has $R^2 \approx 0.95$. The data in the left plot were simulated from the normal linear model.

In short, R^2 should not be reiled upon as a diagnostic to judge the quality of the model fit. **If** the model is a good fit, **then** R^2 says something about the **predictive power** of the model.

⋆ **Exercise:** Suppose that, by mistake, each observed pair (response and predictor) is included twice in the analysis. What is the effect on $\widehat{\beta}_i$, the t-statistics, and R^2?

Diagnostics

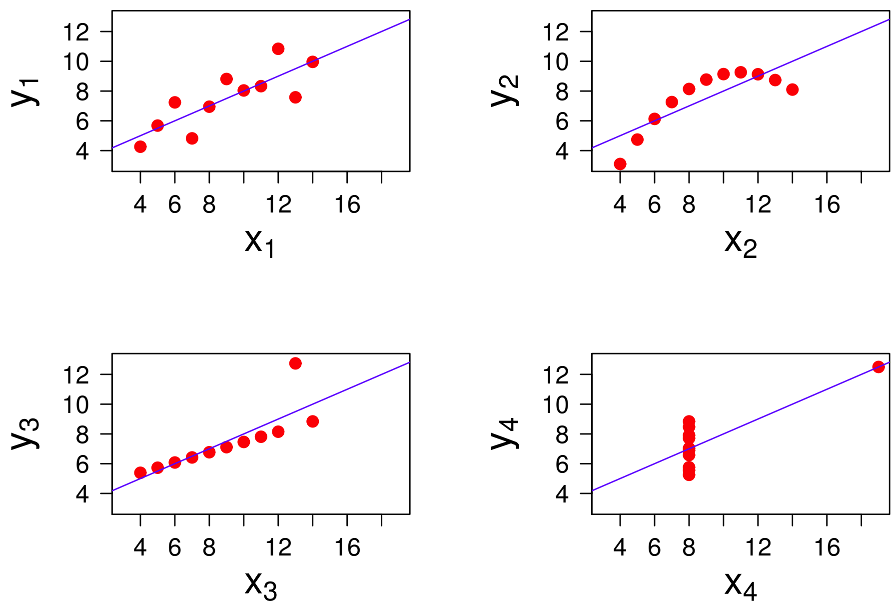


Figure 10: Anscombe's Quartet.

Figure 10 shows four scatter plots constructed by Francis Anscombe for a 1973 research article. These are called **Anscombe's Quartet**.

Exercise: Use Python to find the sample mean and standard deviation for both x and y in all four cases, to find the correlation between x and y in each case, and then use that to find the slope of the regression line for each.

Exercise: This can be taken a step further: We could find that the residual sum of squares in all four cases is exactly the same. What does this imply about the inference for β_1 in all four cases?

Exercise: What is the message we take away from this example?

A very useful diagnostic plot is that of the **residuals versus the fitted values**. There are a wide range of models for which this is an invaluable tool.

Figure 11 shows these plots for the four scatter plots of Figure 10.

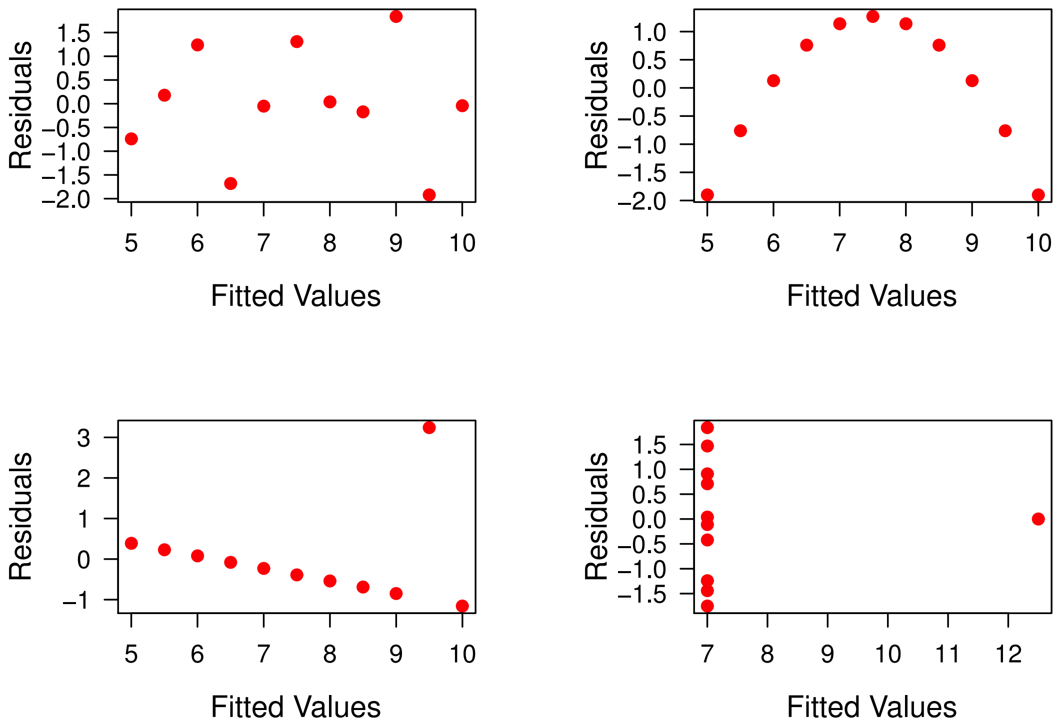


Figure 11: The plots of residuals versus fitted values for Anscombe's Quartet.

⋆ **Exercise:** If the assumptions of the model are correct, what should we see in the plot of residuals versus fitted values?

Exercise: Why is the plot of residuals versus fitted values any more useful than simply looking at the scatter plot?

⋆ **Exercise:** Your colleague says, "I found the correlation between the residuals and fitted values to be equal to zero, so there is clearly no relationship between the residuals and fitted values. So, the model must be fine." Do you agree with this logic?

Heteroskedasticity

One assumption of our model is that all of the model errors have the same variance; this condition is called **homoskedasticity**.

The opposite case, the presence of **heteroskedasticity** is a serious concern. The validity of the confidence intervals, hypothesis tests, and so forth rests on this assumption.

The plot of residuals versus fitted values is a useful tool for diagnosing heteroskedasticity.

Exercise: What shapes would one expect to see in the plot of residuals versus fitted values when heteroskedasticity is present?

There are two major strategies for dealing with heteroskedasticity.

First, the response variable can be **transformed**, meaning you would use $g(y)$ instead of y as the response variable. The two most popular choices for $g()$ are the logarithm, and the square root.

Exercise: Why do you think that the log transform is so often used with financial variables (prices, volume, etc.)?

The other major strategy for dealing with heteroskedasticity is to use **weighted least squares**. Briefly stated, instead of minimizing the residual sum of squares, one minimizes

$$\text{WRSS}(\beta) = \sum_{i=1}^{n} w_i (y_i - \widehat{y}_i)^2$$

where the weights are chosen to deemphasize those observations for which the variance is larger.

In fact, the optimal choice is $w_i = 1/\sigma_i^2$, where σ_i^2 is the variance for the i^{th} model error.

One would not typically know the σ_i^2, but it is sometimes the case that you are willing to assume that $\sigma_i^2 = \sigma^2/m_i$. Then, the weight can be chosen as $w_i = m_i$. Or, the σ_i^2 could be estimated from the residuals.

Exercise: Explain why it may be the case that you would be willing to assume that $\sigma_i^2 = \sigma^2/m_i$.

Example

In this example we will use 82 NYSE stocks in an effort to predict the **realized volatility**, here defined as

$$\frac{1}{n}\sum_{i=1}^{n}\left(\log\left(\frac{P_i}{P_{i-1}}\right)\right)^2$$

where P_i is the closing price on day i.

For this simple model, we will use the logarithm of the total trading volume over the prior month as the predictor of the realized volatility over the upcoming month.

Of course, to **train** or **fit** the model we need data on both the predictor and the response. So, we use two recent consecutive months to form a training set. We will use the aforementioned set of 82 stocks for this purpose.

A simple linear model is fit in attempt to model the relationship between these two variables. Figure 12 shows the raw data and the result of the fit.

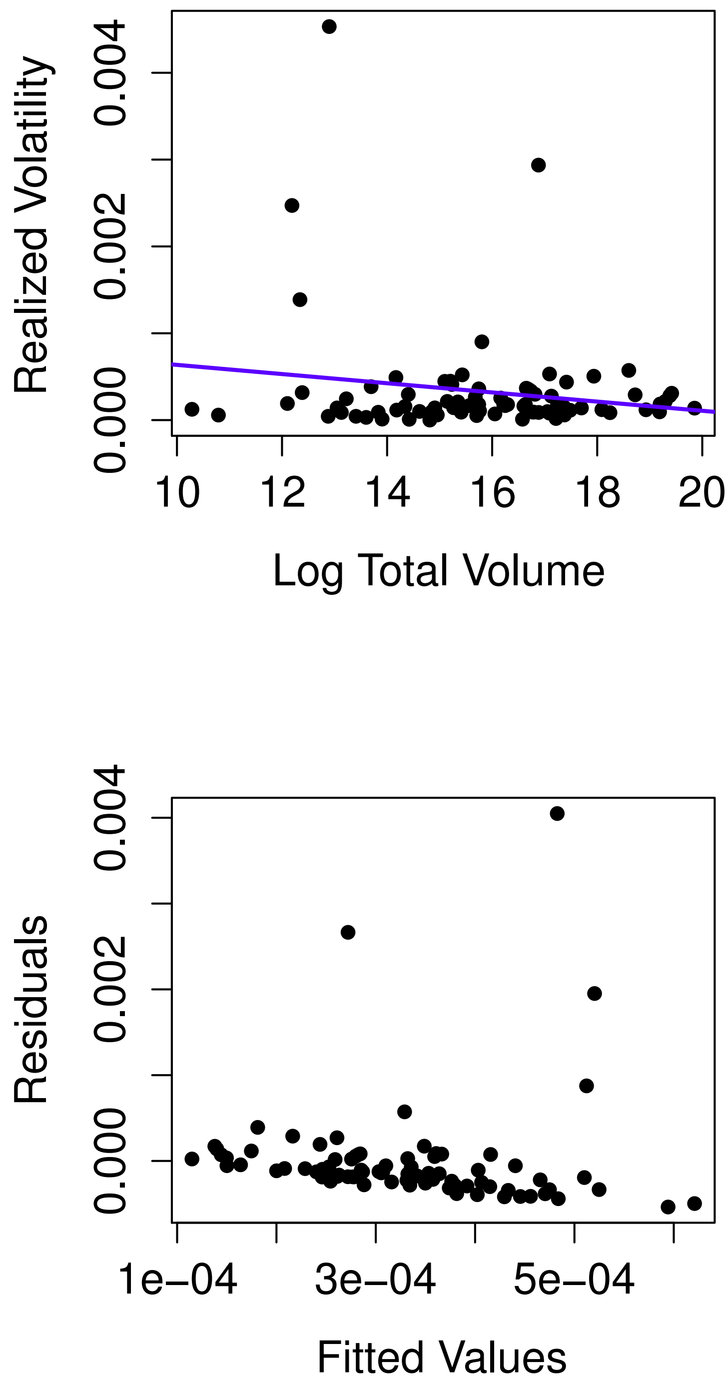
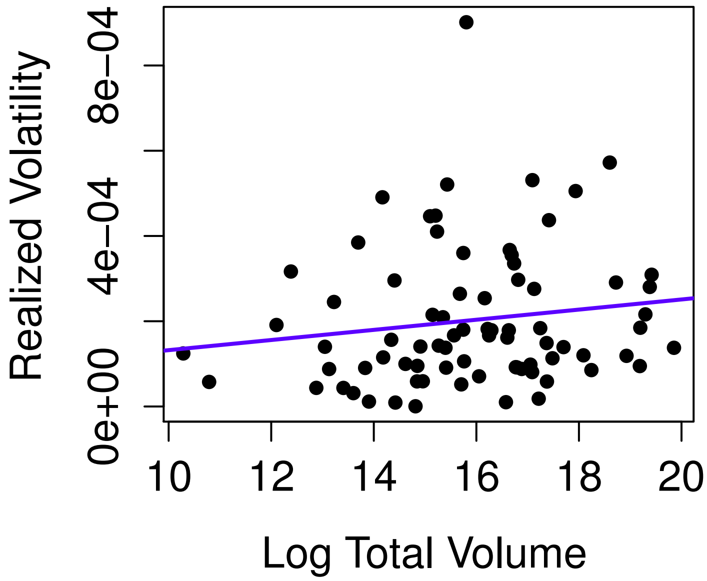


Figure 12: The initial model fit to realized volatility. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

Exercise: Comment on the quality of the fit of this first model.

The model was refit with the four extreme observations removed. Figure 13 shows the result.



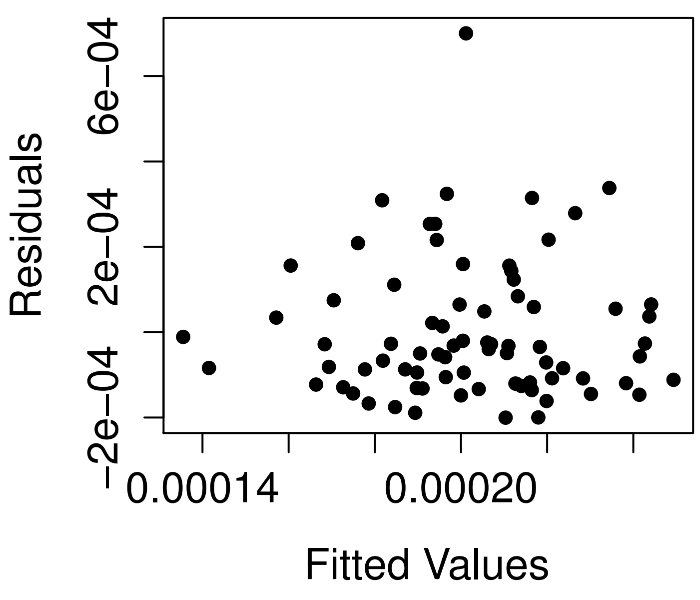


Figure 13: The model fit to realized volatility after the four extreme observations have been removed. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

Exercise: Comment on the quality of the fit of this second model.

The model was then refit with the four extreme observations returned, but using the log transform of the response. Figure 14 shows the result.

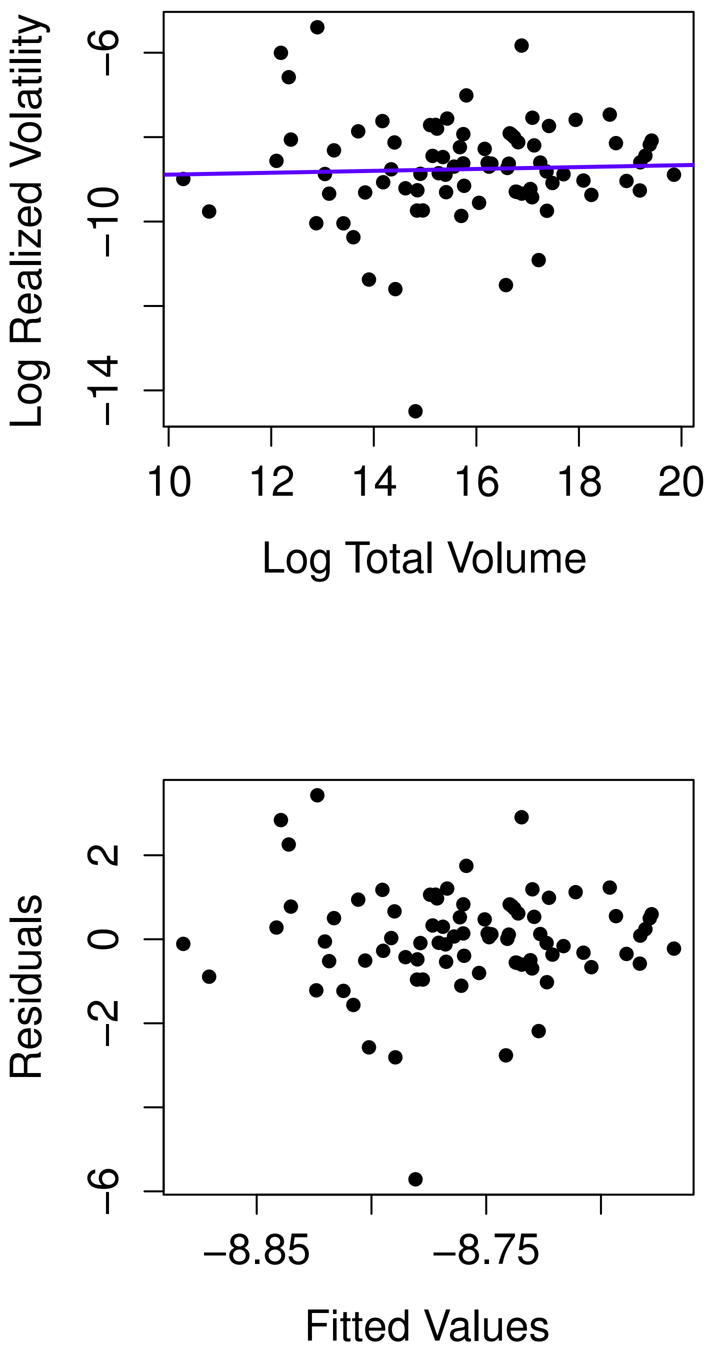


Figure 14: The model fit to realized volatility after taking the log transform. The upper plot shows the scatter plot with the regression line superimposed, the lower plot shows the plot of residuals versus fitted values.

Exercise: Comment on the quality of the fit of this third model.

The Normality Assumption

We should also assess the validity of the assumption that the model errors are normally distributed. This is typically done by looking at a normal probability plot of the residuals.

The distribution of the $\widehat{\beta}_i$ is only exactly normal when the model errors are normal. **But,** there is a version of the central limit theorem that states that $\widehat{\beta}_i$ will be approximately normal as long as the sample size is sufficiently large.

Hence, the standard errors, confidence intervals, and hypothesis tests we construct will still be approximately valid if the model errors are not normal, as long as the sample size is sufficiently large.

Still, the normal probability plot is a valuable tool for identifying **outliers**, which are objects with errors which are much larger than would be expected under the normal distribution.

Also, if we find that the error distribution has heavier tails than would be expected under the normal distribution, we may want to reconsider our use of least squares as the criterion for estimating β.

$\star$ **Exercise:** Explain why least squares may no longer be a good choice if the distribution of the errors was, for instance, the t-distribution with 5 degrees of freedom.

Later we will discuss **robust regression** approaches which use criterion other than least squares.

Figure 15 shows the normal probability plot for the example presented above. This is for the case of the third model (Figure 14).

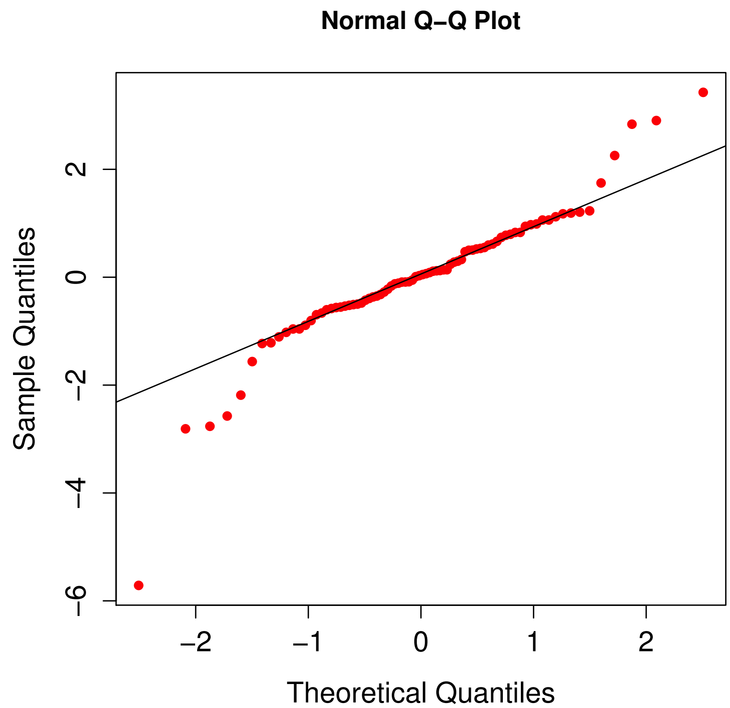


Figure 15: Normal probability plot corresponding to the model shown in Figure 14.

Exercise: Comment on the normal probability plot shown in Figure 15. Is there an issue with outliers? Is the normal assumption valid?

Other Diagnostic Plots

If there is a time dimension to the data that is being used (usually the case with financial data), then one should plot the residuals versus time, and see if there is time-dependence that is not being adequately modelled.

In a subsequent course you will learn about ARMA and GARCH models; these can be built into the assumptions regarding the model error.

It is also a good idea to individually plot the residuals versus each of the predictors in the model. If our model is adequate, there should be no relationship between these quantities.

Making Predictions

One of the fundamental motivations for using regression models is to make predictions for **future** observations of the response. In other words, consider a situation where you know the value of the predictor variable x, and seek to predict the response.

In the context of our simple linear regression model, if our value for the predictor is x^*, then we would start by finding

$$\widehat{y}^* = \widehat{\beta}_0 + \widehat{\beta}_1 x^*$$

We have plugged in this new value for x into the line that has been fit. This is our best estimator for y^*, the true response for this object.

One should not be satisfied with only this, however: We should quantify the amount of error in this prediction.

The **variance in the error in the prediction** is

$$V(\hat{y}^* - y^*) = \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \overline{x})^2}{(n-1)s_x^2} \right]$$

Exercise: Explain how the above decomposes the error in the prediction into two key sources.

Hence the **standard error in the prediction** is

$$\mathrm{SE}(\hat{y}^*) = \sqrt{\sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \overline{x})^2}{(n-1)s_x^2} \right]}$$

Of course, in practice σ^2 will be replaced with $\hat{\sigma}^2$.

Exercise: What happens to the standard error in the prediction as n increases?

A $100(1-\alpha)\%$ **prediction interval for the future response** can be written as

$$\widehat{y}^{*} \pm t_{\alpha/2,n-2}\,\mathrm{SE}(\widehat{y}^{*})$$

In practice, we will rely on software to construct these intervals.

Figure 16 shows a simple example of how the prediction interval will appear, as x^* ranges over the extent of the predictor space.

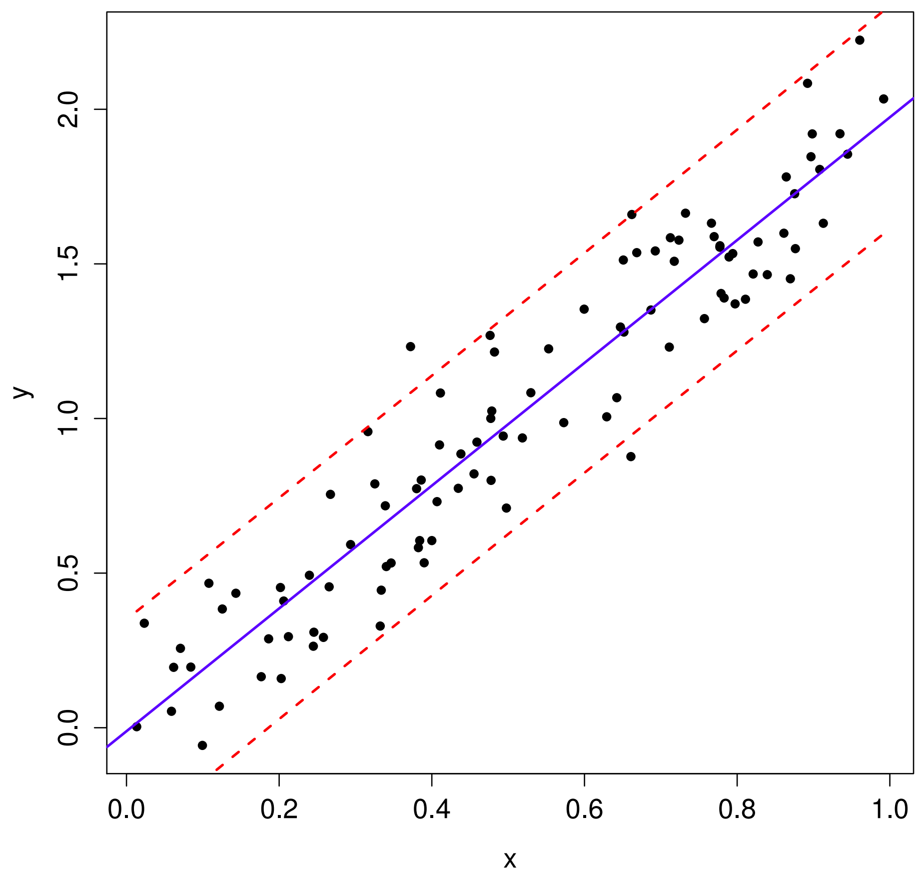


Figure 16: An example showing a regression line (solid line), along with the prediction bands (dashed lines).

Adding a Predictor

Practical regression models typically contain many predictors, i.e., we fit models of the form

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

As more predictors are added to a model, there can be surprising effects on the previously-included predictors:

1. predictors can switch from being "significant" to being "not significant"
2. predictors can switch from being "not significant" to being "significant"
3. the signs on the parameter estimates can flip from negative to positive, and vice-versa

⋆ **Exercise:** Explain why each of the above three things could occur, in the context of going from a model with a single predictor, to a model with two predictors.

When there are two predictors (and the intercept) in a model, the least squares slope estimates can be written as

$$\widehat{\beta}_1 = \left(\frac{r_{y1} - r_{y2}r_{12}}{1 - r_{12}^2}\right)\left(\frac{s_y}{s_1}\right)$$

and

$$\widehat{\beta}_2 = \left(\frac{r_{y2} - r_{y1}r_{12}}{1 - r_{12}^2}\right)\left(\frac{s_y}{s_2}\right)$$

where s_i is the sample standard deviation for predictor i.

⋆ **Exercise:** Consider three regression models:

$$\begin{aligned} Y &= \beta_0 + \beta_1 x_1 + \epsilon \\ Y &= \beta_0' + \beta_2 x_2 + \epsilon \\ Y &= \beta_0'' + \beta_1^* x_1 + \beta_2^* x_2 + \epsilon \end{aligned}$$

Assume parameters are estimated using least squares. What is the relationship between $\widehat{\beta}_1$ and $\widehat{\beta}_1^*$ and between $\widehat{\beta}_2$ and $\widehat{\beta}_2^*$? (It's cleaner to restrict to the case where the response and the predictors have been rescaled to each have standard deviation of one.)